

Adversarial Attack

(x, y_{true}) - object $f_{\theta}(x)$ - NN

$Q(\theta, x, y)$ - loss value

$f_{\theta}(x) = y_{\text{true}}$ $\tilde{x} \leftarrow$ adversarial input

$f_{\theta}(\tilde{x}) \neq y_{\text{true}}$

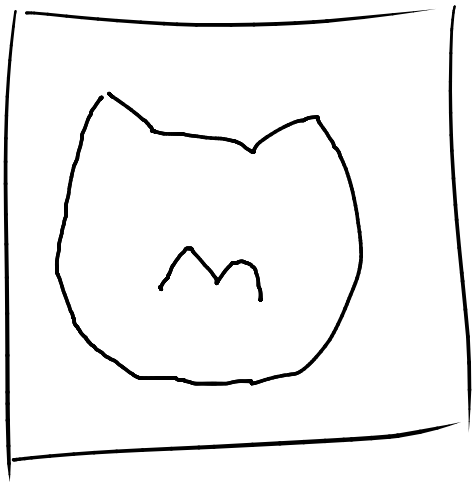
White box - gradient is available

Black box - no gradient

- Targeted attack $f_{\theta}(\tilde{x}) = y_{\text{target}} \neq y_{\text{true}}$
- Untargeted attack $f_{\theta}(\tilde{x}) \neq y_{\text{true}}$

FGSM (Fast Gradient Sign Method)

$$\tilde{x} = x + \epsilon \cdot \text{sign}(\nabla_x Q(\theta, x, y_{\text{true}}))$$



- target

$$\|\tilde{x} - x\|_{\infty} < \frac{\epsilon}{255}$$

$$\max_i |\tilde{x}_i - x_i| \quad \epsilon \sim 1-10 \quad \text{in } 8$$

$$\tilde{x} = x - \epsilon \cdot \text{sign}(\nabla_x Q(\theta, x, y_{\text{target}}))$$

I-FGSM (iterative FGSM)

$$\tilde{x}_0 = x \quad \epsilon, N\text{-iters}$$

for i in $[1, \dots, N]$:

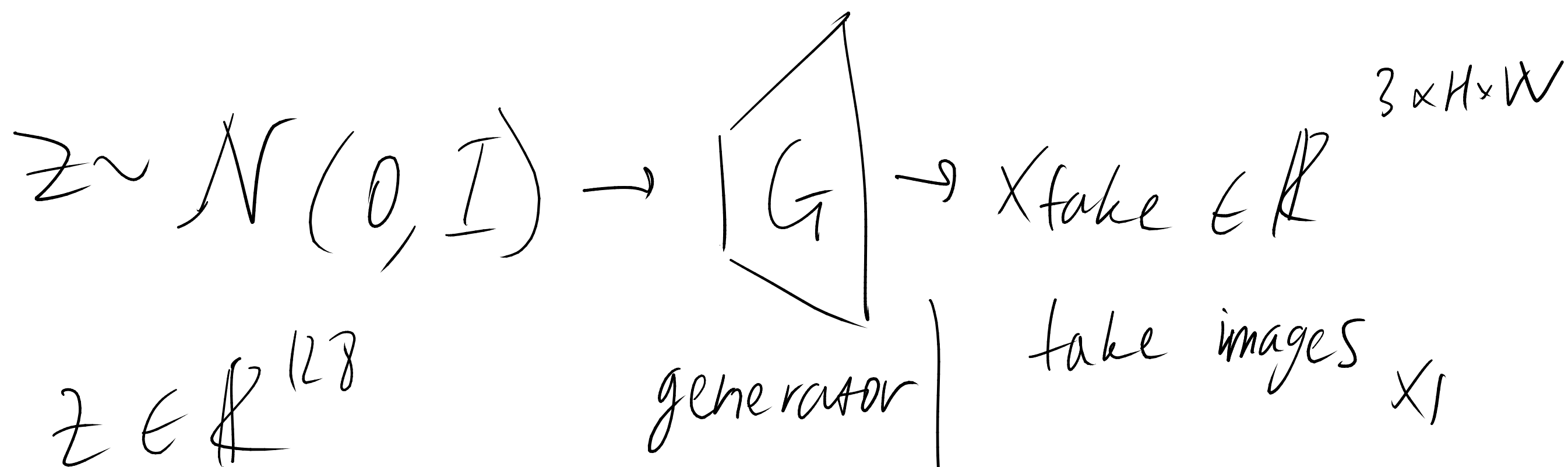
$$\tilde{x}_i = \tilde{x}_{i-1} + \frac{\epsilon}{N} \text{sign}(\nabla_x Q(\theta, \tilde{x}_{i-1}, y_{\text{true}}))$$

$$\leftarrow \tilde{x} = \tilde{x}_N \quad \epsilon \sim 1 \quad \text{in } 8$$

Generative Adversarial Networks (GANs)

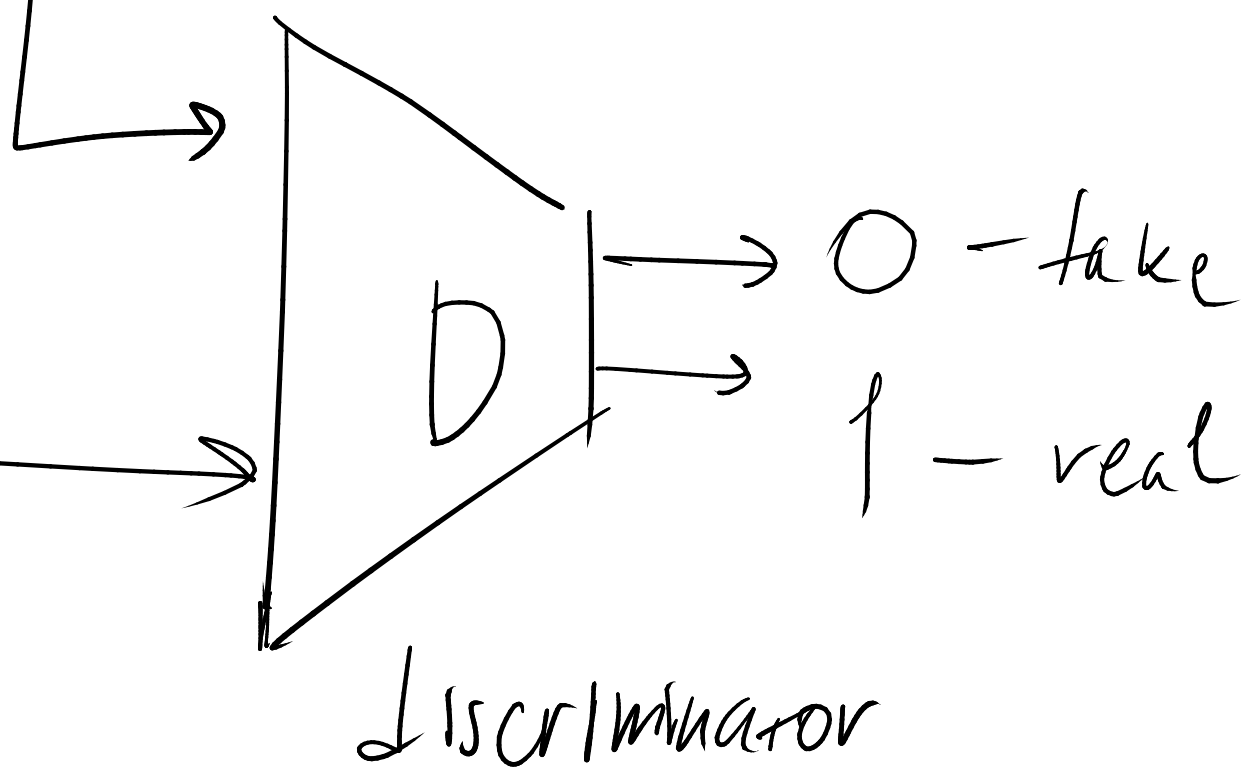
$$\left(x_i \right)_{i=1}^L = X$$

Goal: $x_{\text{fake}} \sim P_{\text{Data}}$



real
Images

x_{real}



$$G: \mathbb{R}^{128} \rightarrow \mathbb{R}^{3 \times H \times W}$$

$$D: \begin{cases} D(G(z)) \rightarrow 0 \\ D(x_{\text{real}}) \rightarrow 1 \end{cases}$$

$$-\log D(x_{\text{real}})$$

$$-\log (1 - D(G(z)))$$

$$\mathbb{E}[-\log D(x_{\text{real}})] + \mathbb{E}[-\log(1 - D(G(z)))]$$

$$x_{\text{real}} \sim p_{\text{Data}}$$

$$z \sim N(0, I)$$

$$\max_G \min_D \left\{ \mathbb{E}[-\log D(x_{\text{real}})] + \mathbb{E}[-\log(1 - D(G(z)))] \right\}$$

$D: \begin{matrix} x_{\text{fake}} \\ \parallel \\ \text{detach}(G(z)) \end{matrix}$
 gen is frozen

Style GAN

Losses:

G : dis is frozen

LSGAN, WGAN, Cramer

$$\max_G \min_D \left(D(x_{\text{real}}) - 1 \right)^2 + D(G(z))^2$$

$$Q_{GAN} = \mathbb{E}_{x_{real} \sim p_{data}} [\log D(x)] + \mathbb{E}_{x_{fake} \sim p_G} [\log (1 - D(x_{fake}))]$$

$\rightarrow \max_D$

$$D^*(x) = ?$$

~~Assume: $p_G = p_{data}$~~

$$Q_{GAN} = \int [p_{data}(x) \log D(x) + p_G(x) \log (1 - D(x))] dx$$

$$x \in \mathcal{X} = \mathbb{R}^{3 \times H \times W}$$

↑
set of all
images

$$g(c) = p_{data}(x) \log c + p_G(x) \log (1 - c)$$

$$g'(c) = \frac{p_{data}(x)}{c} - \frac{p_G(x)}{1 - c} = 0$$

$$(1 - c) p_{data}(x) = c p_G(x)$$

$$D^*(x) = c = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \in [0, 1]$$

$$Q_{GAN} = \int \left[p_{data}(x) \log \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} + p_G(x) \log \frac{p_G(x)}{p_{data}(x) + p_G(x)} + p_{data}(x) \log 2 - p_{data}(x) \log 2 + p_G(x) \log 2 - p_G(x) \log 2 \right] dx =$$

$$= \int \left[p_{data}(x) \log \frac{2p_{data}(x)}{p_{data}(x) + p_G(x)} + p_G(x) \log \frac{2p_G(x)}{p_{data}(x) + p_G(x)} \right] dx -$$

$$- \log 2 \int (p_{data}(x) + p_G(x)) dx =$$

$$\begin{array}{c} x \in \mathcal{X} \\ \parallel \\ 2^{3 \times H \times W} \\ \parallel \\ 2 \end{array}$$

$$= KL \left(p_{data} \parallel \frac{p_{data} + p_G}{2} \right) + KL \left(p_G \parallel \frac{p_{data} + p_G}{2} \right) -$$

$$- 2 \log 2 = 2 \cdot JS(p_{data} \parallel p_G) - 2 \log 2$$

$$\min_G JS(p_{data} || p_G)$$

C GAN conditional

$$G(z, c)$$

c - class label

$$D(G(z, c), c)$$

Head $\begin{matrix} 0 & \text{fake} \\ 1 & \text{real} \end{matrix}$

Head

classification on C

$$D(x_{real}, c)$$

Pix2Pix

$$\{(x, y)\}$$

x - satellite image

y - map

Facades, Sat2Map, Super resolution

$$G(x)$$

$$D(G(x)) \rightarrow \begin{matrix} 0 \\ 1 \end{matrix}$$

$$MSE(G(x), y)$$

$$\alpha \text{MSE} + (1-\alpha) \text{D}_{\text{loss}}$$